

9 Regularization

$$9.1 \quad P(\phi) = \prod_{j=1}^J \text{Norm}_{\phi_j} [0, \sigma_\phi^2]$$

$$\hat{\phi} = \underset{\phi}{\operatorname{argmax}} \prod_{i=1}^I P_r(y_i | x_i, \phi) P_r(\phi)$$

$$= \underset{\phi}{\operatorname{argmin}} -(\log [\prod_{i=1}^I P_r(y_i | x_i, \phi) P_r(\phi)])$$

$$= \underset{\phi}{\operatorname{argmin}} -(\sum_{i=1}^I \log [P_r(y_i | x_i, \phi)] + \log [P_r(\phi)])$$

$$\text{As } \log P_r(\phi) = \log \left(\prod_{j=1}^J \frac{1}{\sqrt{2\pi\sigma_\phi^2}} \exp \left[-\frac{\phi_j^2}{2\sigma_\phi^2} \right] \right)$$

$$\propto -\frac{1}{2\sigma_\phi^2} \sum_{j=1}^J \phi_j^2$$

$$\text{This simplifies to } \underset{\phi}{\operatorname{argmin}} \underbrace{\sum_{i=1}^I -\log [P_r(y_i | x_i, \phi)]}_{=: \ell_i[x_i, y_i]} + \underbrace{\frac{1}{2\sigma_\phi^2} \sum_{j=1}^J \phi_j^2}_{=: \lambda} \quad \square$$

9.2 All derivatives w.r.t. ϕ_j have a $\lambda \phi_j$ added to them:

$$\frac{\partial \mathcal{L}(\phi)}{\partial \phi_j} + \lambda \phi_j \quad \text{as it's linear in parameter!}$$

Q.3 $y = \phi_0 + \phi_1 x$

$$L(\phi) = \sum_{i=1}^I (\phi_0 + \phi_1(x_i + \epsilon) - y_i)^2$$

$$= \sum_{i=1}^I (\phi_0 + \phi_1 x_i - y_i + \phi_1 \epsilon)^2$$

$$= \sum_{i=1}^I \left((\phi_0 + \phi_1 x_i - y_i)^2 + 2(\phi_0 + \phi_1 x_i - y_i)(\phi_1 \epsilon) + (\phi_1 \epsilon)^2 \right)$$

$$E(L(\phi)) = \sum_{i=1}^I (\phi_0 + \phi_1 x_i - y_i)^2 + 0 + \phi_1^2 \sum_{i=1}^I \overbrace{E(\epsilon^2)}^{\sigma_x^2}$$

as $E(\epsilon) = 0$, and, as ϵ independent, $E(\epsilon^2) = \text{Var}(\epsilon) = \sigma_x^2$.

This is equivalent to L2 regularization with $\lambda = \sigma_x^2 / 6$

Q.4 we replace y with $y_y = 0.9$ and $y_e = \frac{0.1}{D_0 - 1}$:

$$L(\phi) = - \sum_{i=1}^I \log \left[0.9 \text{softmax}_k [f(x_i, \phi)] + \sum_k \frac{0.1}{D_0 - 1} \text{softmax}_k [f(x_i, \phi)] \right]$$

9.5

or $\phi \leftarrow (1-\lambda)\phi - \alpha \frac{\partial L}{\partial \phi}$ will decay rate λ

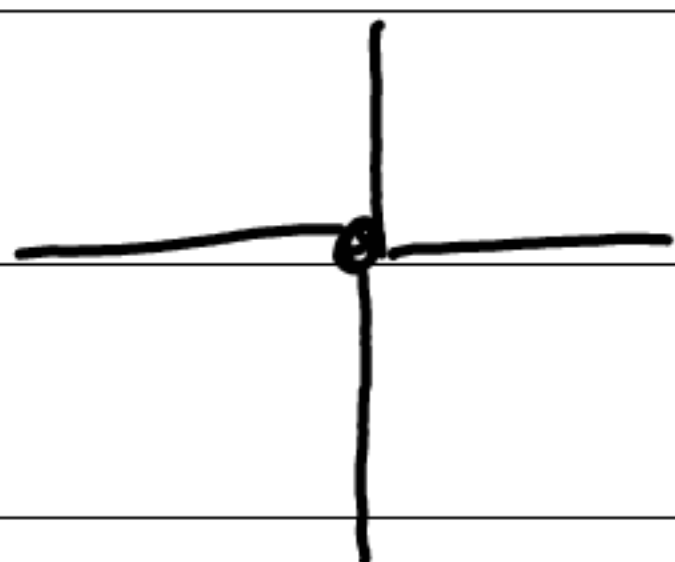
$$\Rightarrow \tilde{L}[\phi] = L[\phi] + \frac{\lambda}{2\alpha} \sum_k \phi_k^2$$

i.e., a weight decay step is equivalent to a GD step on $L + \frac{\lambda}{2}$

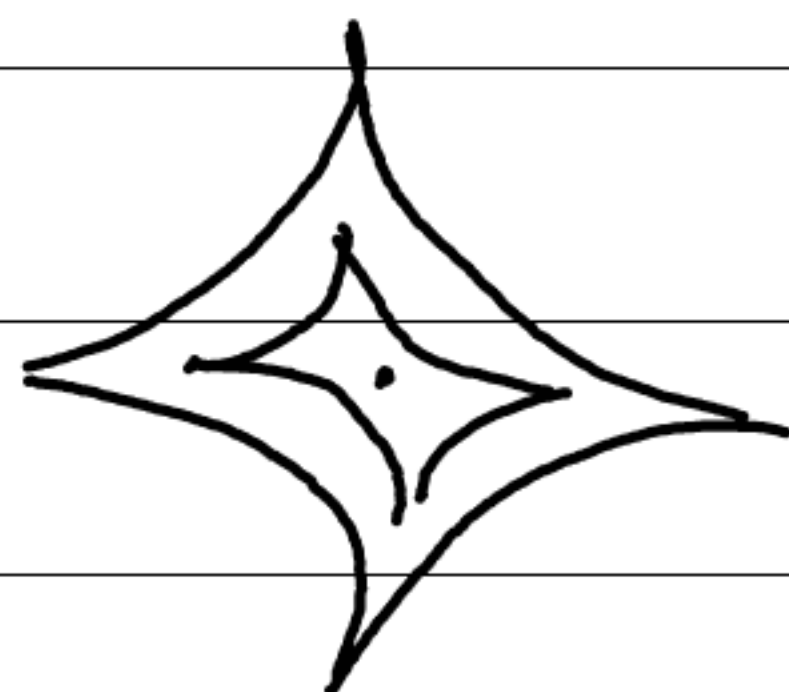
$$\nabla_{\phi} \tilde{L} = \nabla_{\phi} L + \frac{\lambda}{2\alpha} \nabla_{\phi} \sum_k \phi_k^2 = \nabla_{\phi} L + \frac{\lambda}{\alpha} \phi$$

$$\begin{aligned} \text{as } \frac{\partial \phi_k^2}{\partial \phi_k} &= 2\phi_k. \text{ As } \phi \leftarrow \phi - \alpha \nabla_{\phi} \tilde{L} \\ &= \phi - \alpha \left(\nabla_{\phi} L + \frac{\lambda}{\alpha} \phi \right) \\ &= \phi - \alpha \nabla_{\phi} L - \lambda \phi \\ &= \phi - \lambda \phi - \alpha \nabla_{\phi} L \\ &= (1-\lambda)\phi - \alpha \nabla_{\phi} L \quad \square \end{aligned}$$

9.6 LO:



$L_{1/2}$:



L_1 :

